

Using Large Generative Models for Storyboarding: Challenges and Goals

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CCS CONCEPTS

• **Human-centered computing** → **User interface programming**.

KEYWORDS

Storyboarding, creative tool, large generative models

1 INTRODUCTION

Having a good idea does not usually mean having good video production. In order to bridge the gap between ideation and final video production, video makers often generate an intermediary product known as a storyboard to streamline the creative process [3].

A storyboard usually contains a script and the visual representations of each line of words that are formatted in a two-column style. Typically, the left column contains timestamped textual descriptions of various elements, such as dialogue and notes on scene changes and background music genre. The right column features a visual depiction of what the scene in the video may look like corresponding to each line of the script.

Storyboard is a common practice among video makers due to several merits. Firstly, storyboarding facilitates planning by enabling video makers to strategically sketch out each shot in advance. Consequently, video makers can visualize how the scenes will flow together and identify potential issues or gaps in the narrative. Secondly, storyboard can facilitate the production process, resulting in time and cost savings. Thirdly, it empowers video makers to exercise creative control over the visual elements of the video by flexibly experimenting with different shots, angles, and compositions to achieve the desired visual effect. Therefore, it has been widely adopted among video makers, especially those who have stringent requirements for the quality of the videos [7].

However, producing a storyboard can be tedious and time-consuming due to several factors. Firstly, writing the script itself is a challenging process, as the writer needs to transfer their high-level comprehension of the story into a structured, concrete, and coherent form. Additionally, the created visual plots has to consistently align with the written script, which requires certain expertise from the producers. Meanwhile, since the storyboard is an intermediary product for producing a video, it requires specific annotations such as camera angles and music genres at a specific time.

There have been notable advances in the field of large generative models (LGMs) such as GPT-3 [1] and Stable Diffusion [5] lately. These models have shown the ability to generate high-quality content in some cases, such as [2, 4, 6, 9]. Given the successful application of LGMs in these domains and their vast potential, it is worthwhile to explore how they can be used in a more complex

multimodal domain. In this position paper, we focus on discussing the challenges and goals of LGM-assisted storyboarding.

2 DESIGN CHALLENGES

DC1: How to generate consistent characters and scenes in a video script. This challenge stands out particularly when generating long content, as the prompts that drive the large language model, the interplay between various characters and scenes, and the connections between these elements become more complicated. Failure to maintain consistency in these aspects can lead to a fragmented storyline.

DC2: How to establish the correspondence between the script and visual elements both at a specific moment and throughout the storyboard. Storyboarding requires generating content in two modalities (i.e. script and image), at specific timestamps. However, current LGMs can only generate content unidirectionally, i.e., text to image or image to text. Therefore, multiple models are required to work together, which makes the result more susceptible to errors. Furthermore, as discussed in DC1, a storyboard has to maintain continuity over time, the same principle also applies to the correspondence between script-visual element pairs. To illustrate, when the script points to a visual scene that has been generated before, the model should stringently generate the same scene or minor variations to that scene.

DC3: How to personalize LGMs to fit a particular creator. In the domain of storyboard composition, the process is highly personalized. Creators need to have the narrative reflect their distinct styles and preferences rather than producing generic and commonplace content that can be directly generated by models.

3 DESIGN GOALS

DG1: Establishing inter- and intra- modality connections in the storyboard. Specifically, inter-modality connection pertains to the relationship between the textual script and the visual elements, while intra-modality connection involves the temporal coherence within the textual script and the sequence of visual representations. To realize this design objective, one possible strategy could be designing a semi-structured chain of prompts for both the text and image generation model. The stable part of the chain ensures consistency in the generated output, while the varied section enhances the adaptability and learning capacity of the AI assistant in response to user input.

DG2: Enabling effective human-AI interaction for intent clarifying and error handling. At the intent level, since the story plot varies over time, users need to frequently *communicate* with the AI agent to ensure that it accurately captures their evolving intentions. Therefore, our design goal is to devise an iterative

mechanism, allowing both the user and AI to comprehend each other's state step-by-step [8], progressing from vague to concrete. At the error handling level, an effective interaction strategy should be devised to minimize the user's effort when editing the content at a particular place and to facilitate broadcasting the local edits to other places.

DG3: Balancing creators' reliance on AI and their past domain knowledge and experience in making storyboards. As storyboard is more frequently adopted by professional users who have developed habits and had experience in creating storyboards in their own style, our system should present a user-friendly workflow that aligns with the user's existing workflow for drafting the storyboard.

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